

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set(style="darkgrid")
```

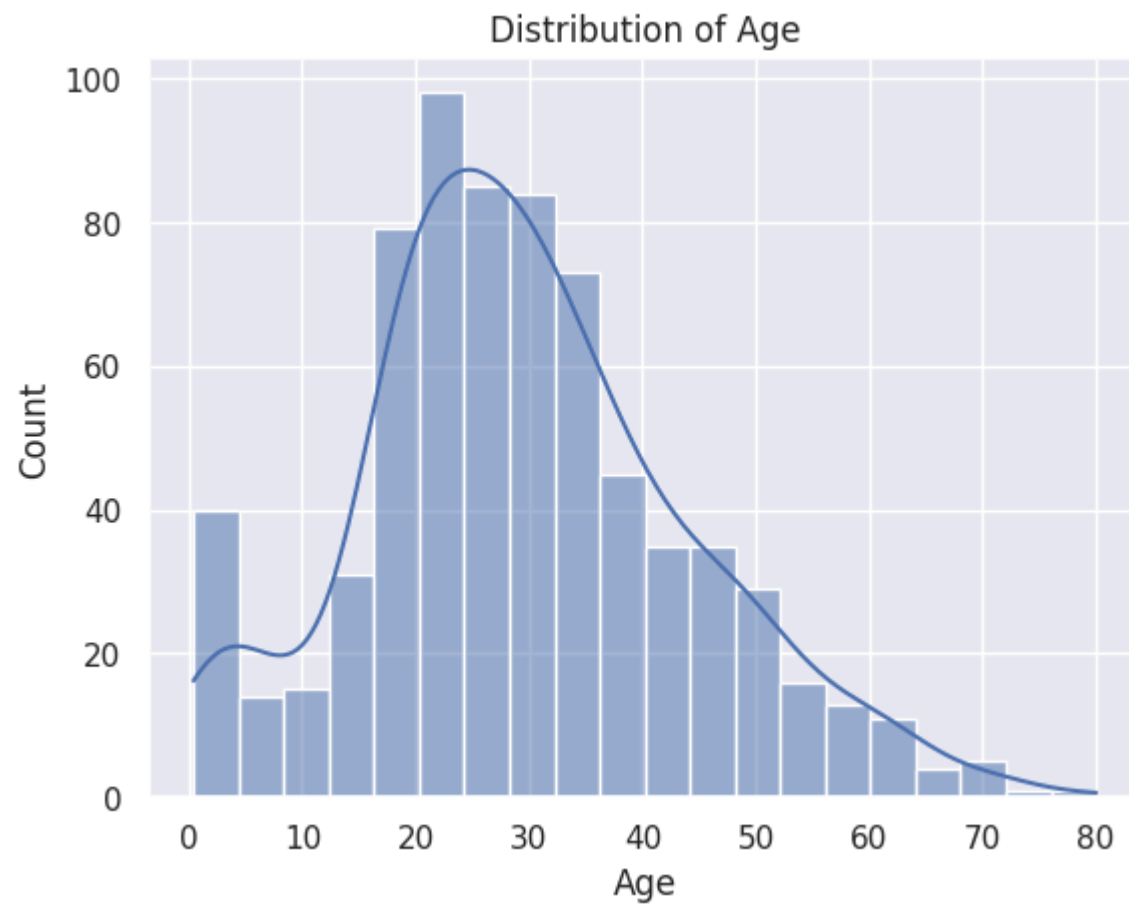
```
df = pd.read_csv('Titanic-Dataset.csv')
```

```
df.head()
df.info()
df.describe(include='all')
df.isnull().sum()
df.shape
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   PassengerId 891 non-null    int64
 1   Survived    891 non-null    int64
 2   Pclass      891 non-null    int64
 3   Name        891 non-null    object
 4   Sex         891 non-null    object
 5   Age         714 non-null    float64
 6   SibSp       891 non-null    int64
 7   Parch       891 non-null    int64
 8   Ticket      891 non-null    object
 9   Fare        891 non-null    float64
10   Cabin       204 non-null    object
11   Embarked    889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
(891, 12)
```

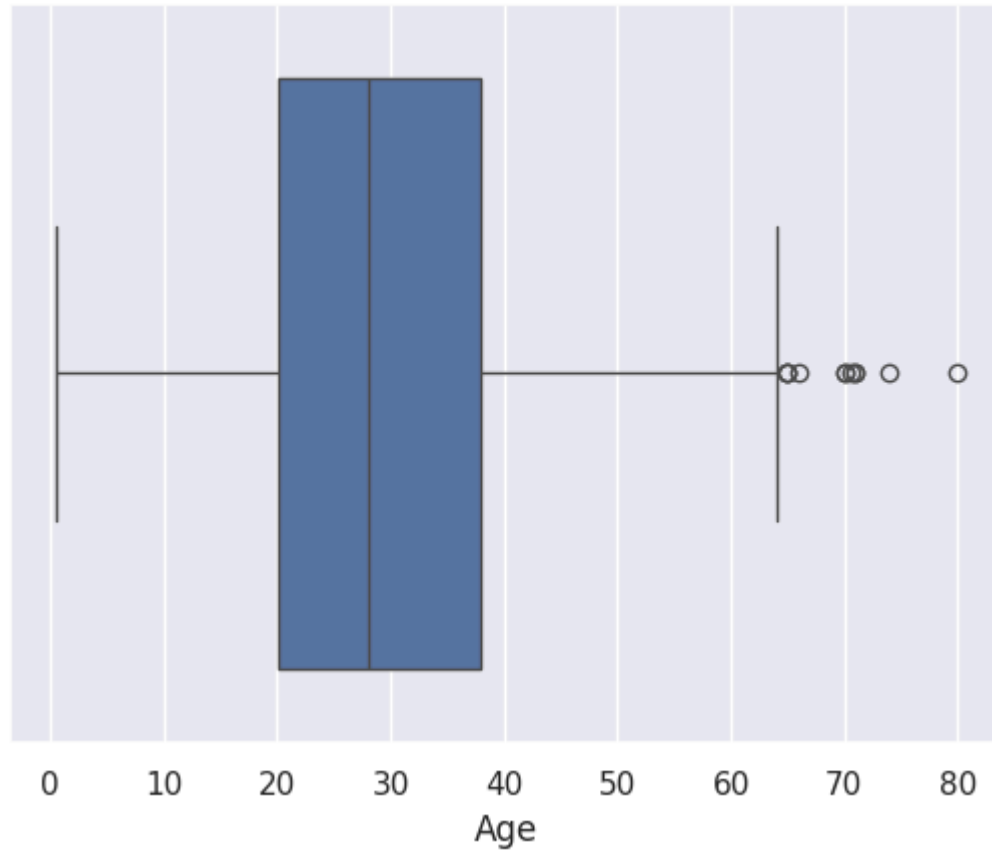
```
sns.histplot(df['Age'].dropna(), kde=True)
plt.title('Distribution of Age')
plt.show()
```



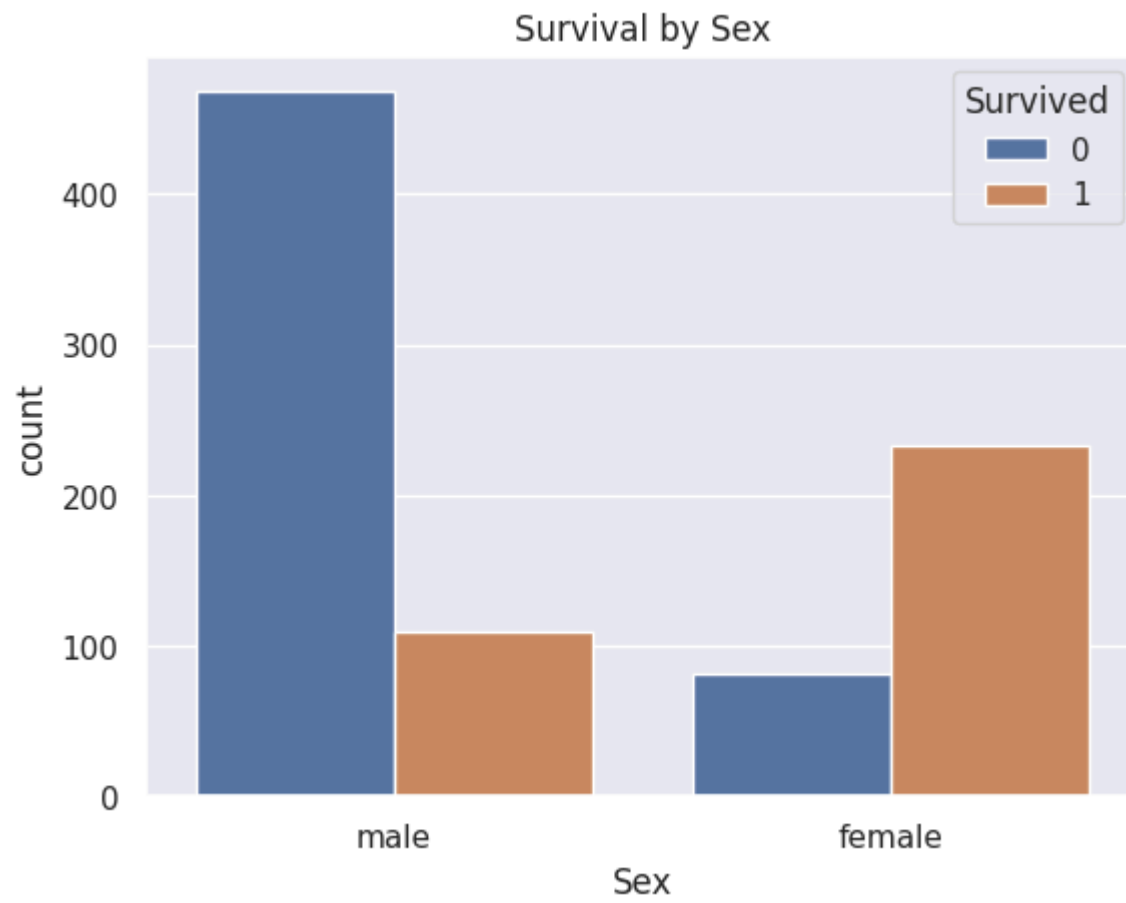
```
sns.boxplot(x='Age', data=df)
plt.title('Boxplot of Age')
plt.show()
```



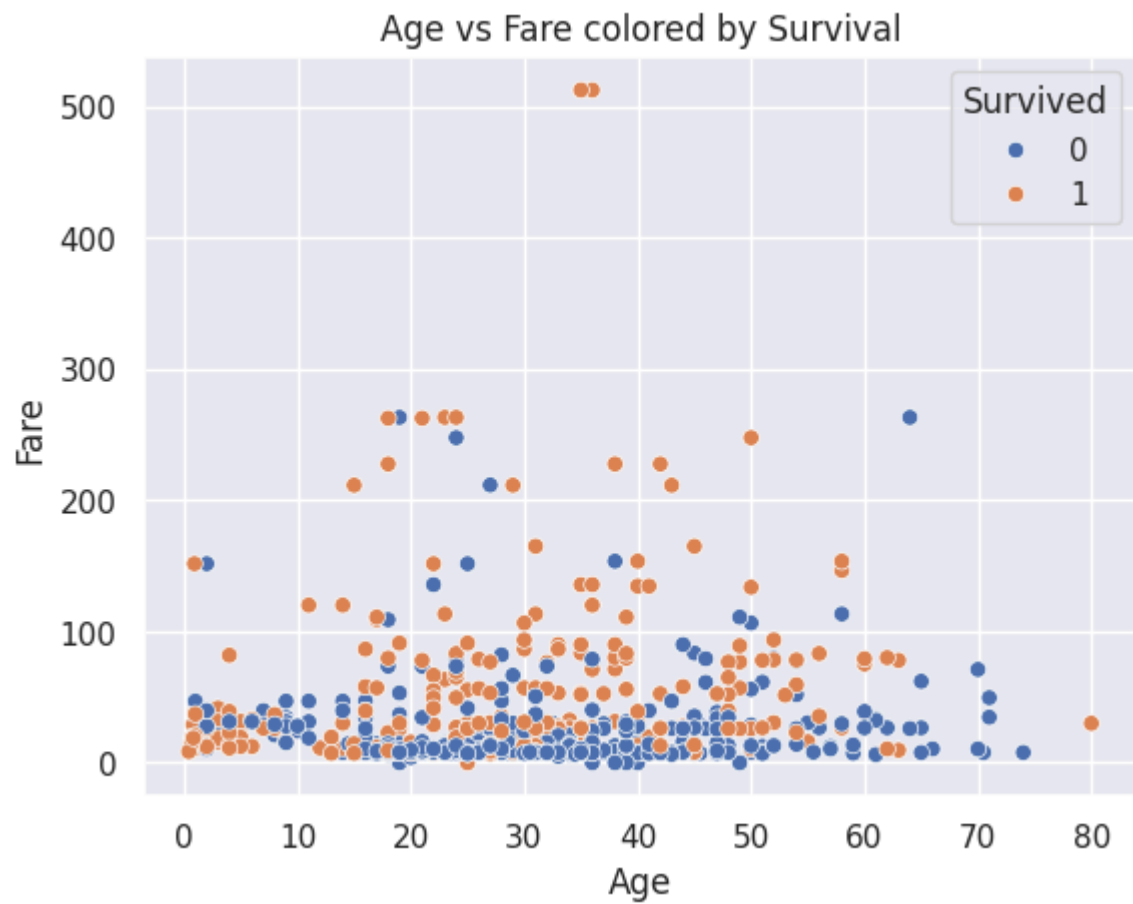
Boxplot of Age



```
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Sex')
plt.show()
```



```
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title('Age vs Fare colored by Survival')
plt.show()
```

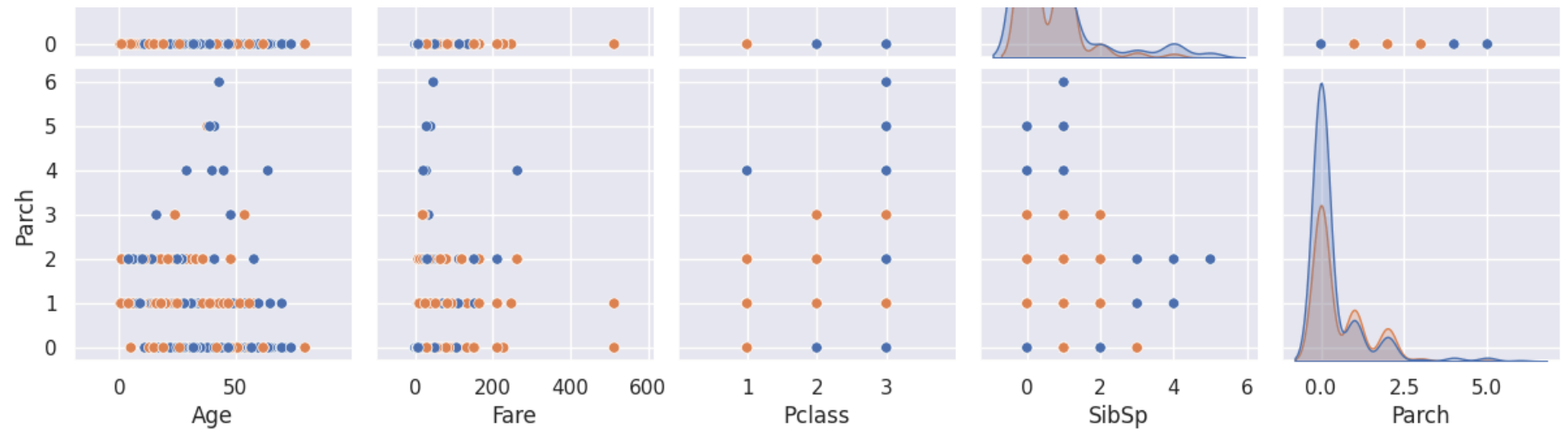


```
sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass', 'SibSp', 'Parch']].dropna().astype(float), hue='Survived')  
plt.suptitle('Pairplot of Numerical Features', y=1.02)  
plt.show()
```



Pairplot of Numerical Features





```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

